Using Data Analysis to Address Important Social Questions

and now on to thinking about a social issue …

Consider this Statement:

"Obesity reduces health in the community"

Hypotheses:

Ha: Higher levels of obesity reduce the average level of health. (this is what we think based on our current knowledge of obesity and health.)

Knowledge gained from where?

Ho: Higher levels of obesity does not change the average level of health.

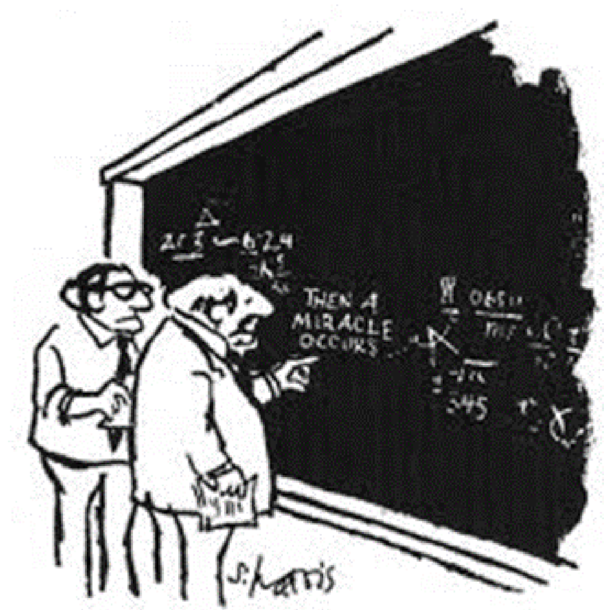
Which hypothesis(s) do we Test and Why?

Think about how we can test this hypothesis:

Ho: Higher levels of obesity does not change the average level of health

Framework for consideration you may choose to apply:

- Who
- What
- When
- Where;
- Why, &;
- How…..



"I think you should be more explicit here in step two."

What phenomenon are we examining?

Ho: Higher levels of obesity does not change the average level of health

What phenomenon of interest?

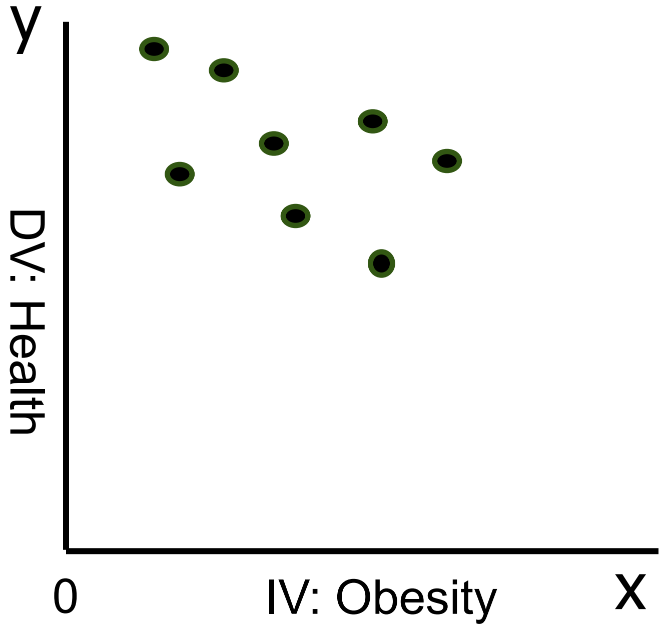
Health (dependent variable; *y*)

What influences that phenomenon?

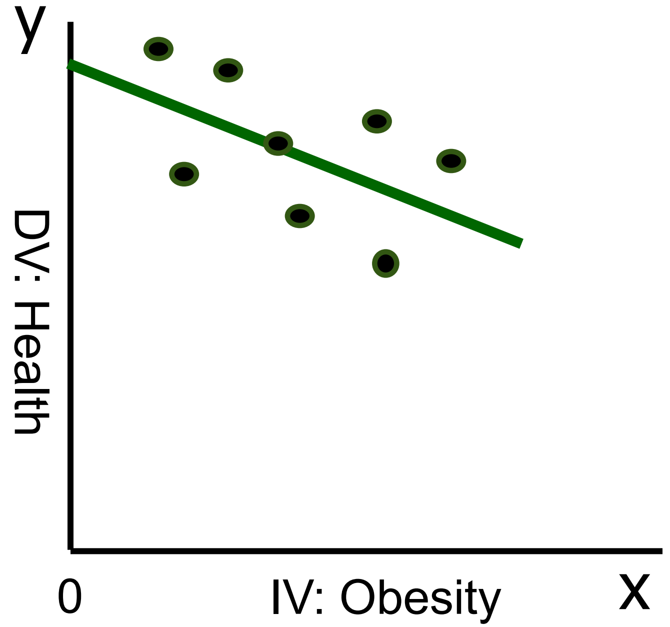
1. Obesity (independent variable; x)

Task: Visualise this relationship and draw what you think it looks like

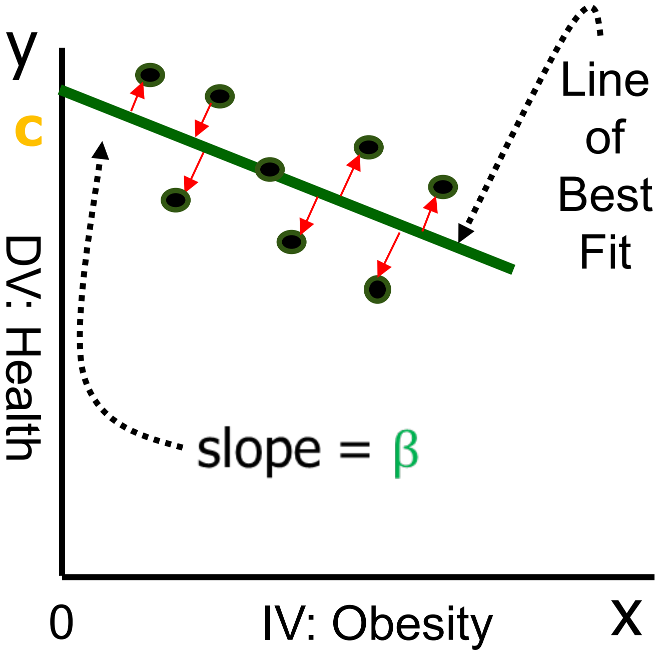
Relationship model



Relationship model



Relationship model



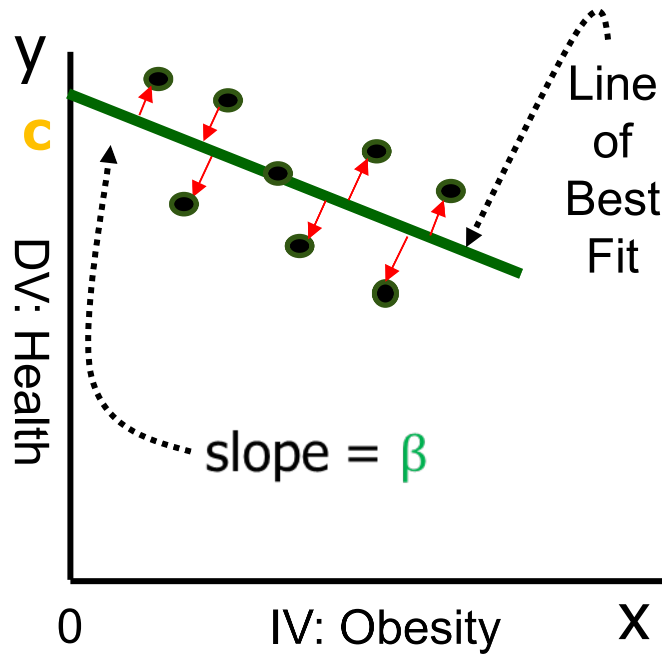
Relationship model

c the intercept; the value of y when x = 0

β the slope of the line; a measure of the rate of change ($\frac{dy}{dx}$) in y for a unit change in x

ε that part of health that obesity does not explain; error term

Line of Best Fit: $$min \sum_{n=1}^{N} (\varepsilon^2)$$



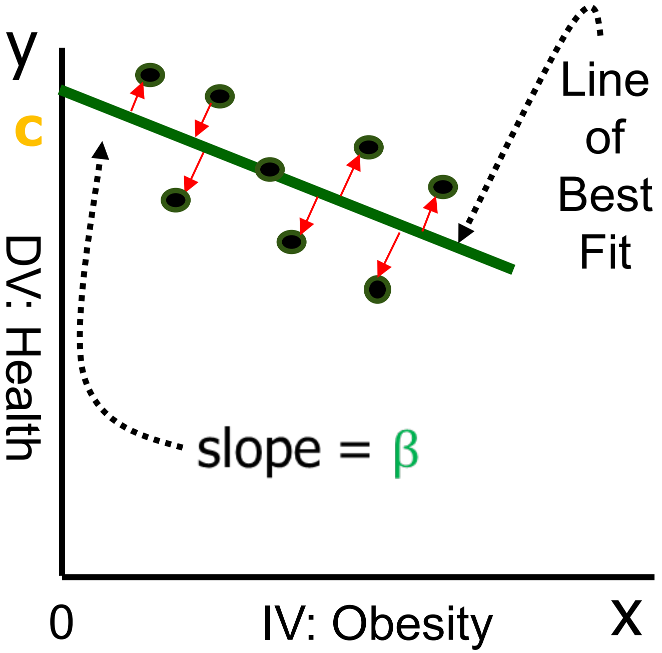
Linear Relationship model

c the intercept; the value of y when x = 0

β the slope of the line; a measure of the rate of change ($\frac{dy}{dx}$) in y for a unit change in x

Ɛ that part of health that obesity does not explain; error term

Line of Best Fit: $min \sum_{n=1}^{N}(\varepsilon^2)$



$$Equation\ of\ the\ line:\ y{=}f(x)\ ;\ y = c - \beta x + \varepsilon\ ; \{x1, x2, x3 \ldots .xN\}$$

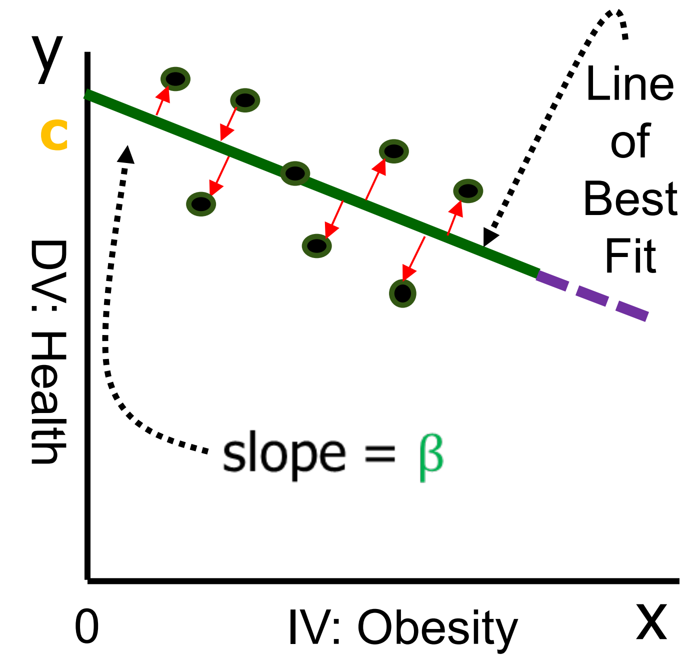
Linear model (Ordinary Least Squares)

c the intercept; the value of y when x = 0

β the slope of the line; a measure of the rate of change ($\frac{dy}{dx}$) in y for a unit change in x

Ɛ that part of health that obesity does not explain; error term

Line of Best Fit: $min \sum_{n=1}^{N}(\varepsilon^2)$



$$Equation\ of\ the\ line:\ y{=}f(x)\ ;\ y = c - \beta x + Ɛ\ ; \{x1, x2, x3 \ldots . xN\}$$

Prediction: If we do not have data past a certain value of obesity (say 30), how can we predict the change in average health (y) if obesity (x) was to rise to 32?

Apply the equation of the line: β= -0.4; c=4.5: 4.5 +(-0.04*32)=3.22

Task: Predict health when obesity=50

What else has an effect on Health?

Ho: Higher levels of obesity does not change the average level of health

What phenomenon of interest?

Health (dependent variable; *y*)

What influences that phenomenon?

1. Obesity (independent variable; x)
2. Etc.
3. Etc.

What else could have an effect on health? (control variables)

Task: What else has an effect on Health?

Hint: How can I find out what else has an effect on health?

Your List of variables?

Time to collect some data

Ho: Higher levels of obesity does not change the average level of health

For Who: Australians; the population for our study.

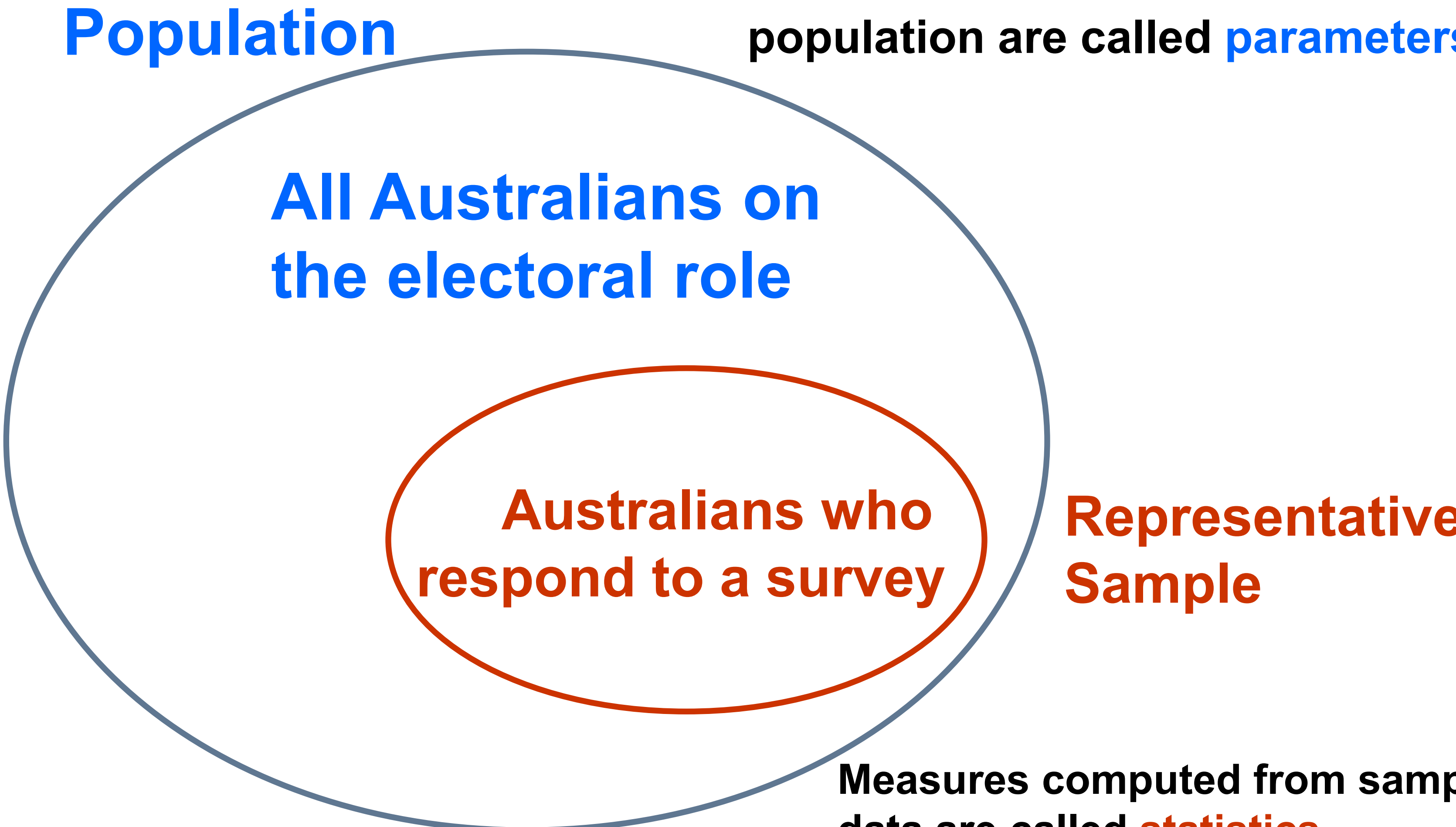
Task: How would you rewrite the hypotheses now that we have decided to look at Australians?

Ha: Higher levels of obesity in Australians reduces the average level of health of Australians.

Ho: Higher levels of obesity in Australians does not change the average level of health of Australians.

Population v Sample

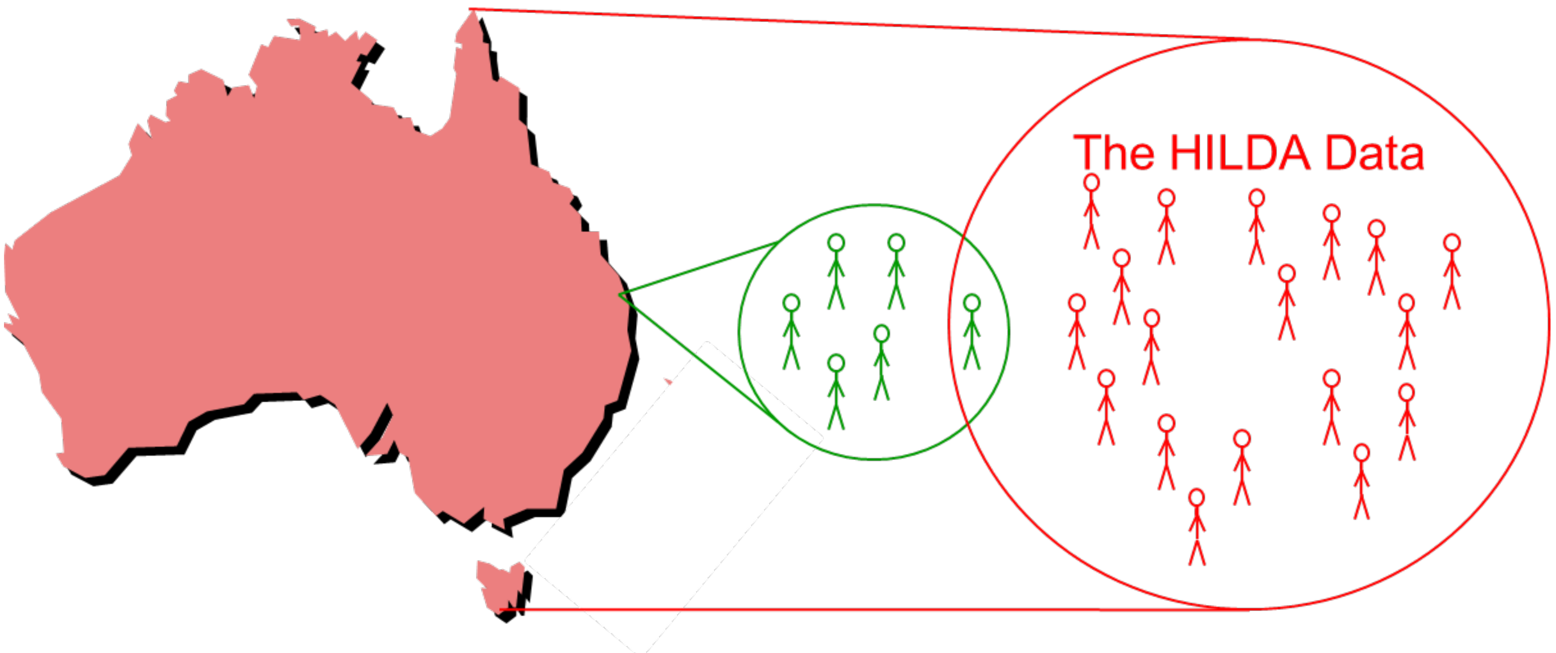
Measures used to describe a population are called parameters



Measures computed from sample data are called statistics

Collect Data from Who?

- **WHO** are we interested in (Australian population)?
- Do we want to collect data from Australian: Individuals (*i*), groups (*g*)? (unit of analysis)?
- Which people in the Australian population do we ask to complete a survey (random sampling)?



[Example: HILDA for Data Users (unimelb.edu.au)]()

Time to collect data

- At one point in time (cross-sectional data;
- Over a Number of Periods (t) (time-series data);
- Pooled, a combination of Time Series and Cross-sectional, or;
- Something else?

Task: How often do we want to collect the data and why?

Types of Data

- **What** type of data would inform us about the DV?
- **What** type of data would inform us about the IVs?

Task 4: Consider the above

Time to collect data

What type of data would inform us about the DV?

Health: self-reported general health factor from the SF-36 international health self-reported health question:

PART A: GENERAL HEALTH AND WELL-BEING
(SF-36 Health Survey)

This first set of questions seeks your views about your health, how you feel and how well you are able to do your usual activities.

Please take the time to read and answer each question carefully by crossing the box corresponding to your response. If you are unsure about how to answer a question, please give the best answer you can.

A1 In general, would you say your health is:

(Cross ☒ ONE box)

☐ Excellent (1)	☐ Very good (2)	☐ Good (3)	☐ Fair (4)	☐ Poor (5)	sgh1

Scale: 1 (min) to 5 (max); nominal, discrete (1,2,3,4,5); range = 5; reverse coded; rescale

Is the scale Cardinal, or;

Ordinal?

0 1 2 3 4 5

0 1 2 3 4 5

Task: What difference would ordinal/cardinal data make to your visualisation of the obesity - health relationship?

Time to collect data

What type of data would inform us about the IV?

Obesity: Body Mass Index (BMI); *f (self-reported weight [kg]; height [metres$^{2]}$)*

B8 How tall are you (without shoes)?
You only need to provide an answer in either centimetres (cms) or in feet / inches.

☐☐☐ *cms* **OR** ☐ *feet* ☐☐ *inches*
(Note: There are 12 inches in a foot)

sbmhtcm
sbmhtft
sbmhtin

B9 What is your current weight?
You only need to provide an answer in either kilograms (kgs) or in stones / pounds.

☐☐☐ *kgs* **OR** ☐☐ *stones* ☐☐☐ *pounds*
(Note: There are 14 pounds in a stone)

sbmwtkg
sbmwtst
sbmwtlb

Scale (kg): numeric, continuous; range $^{0 = \infty}$

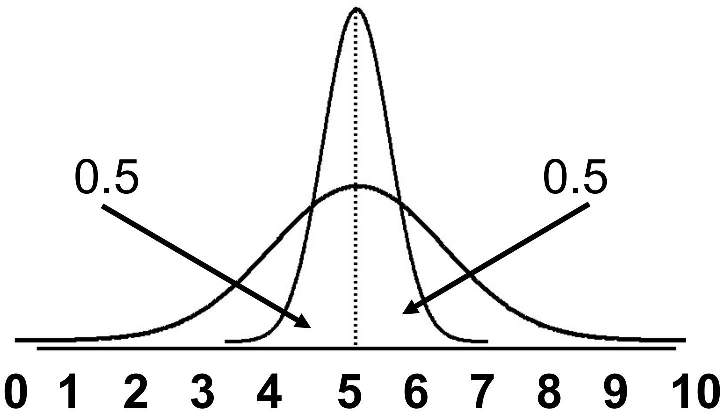
$$\text{BMI} = \frac{weight\ (kg)}{height\ (m^2)}$$

Task: Consider the accuracy of self-reported height and weight, and the constructed BMI factor?

Other types of data sources & scales?

Task: Consider other DVs; IVs; data types and scales; list them.

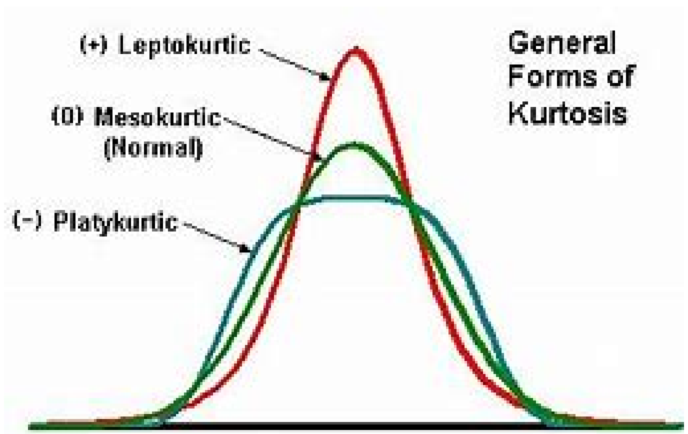
Normal Distribution of a Variable



Task: What happens to the Distribution of a variable as the Sample Size changes?

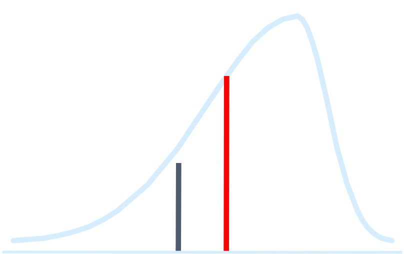
Shape of a Distribution

Measures of shape describe how data are distributed: symmetric or skewed; kurtotic; outliers



Left-Skewed

Mean < Median

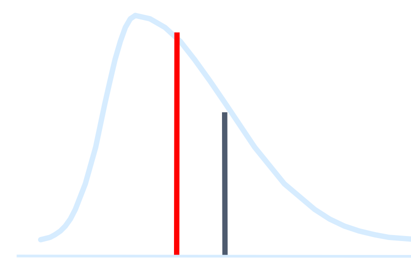


Symmetric

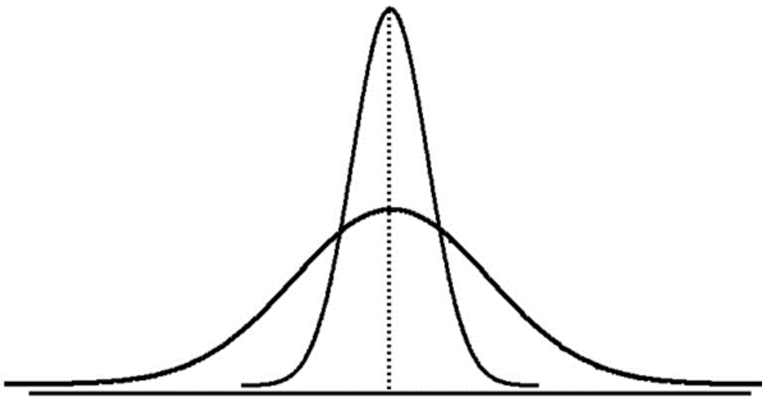
Mean = Median

Right-Skewed

Median < Mean



Distribution of the Health & BMI Variables

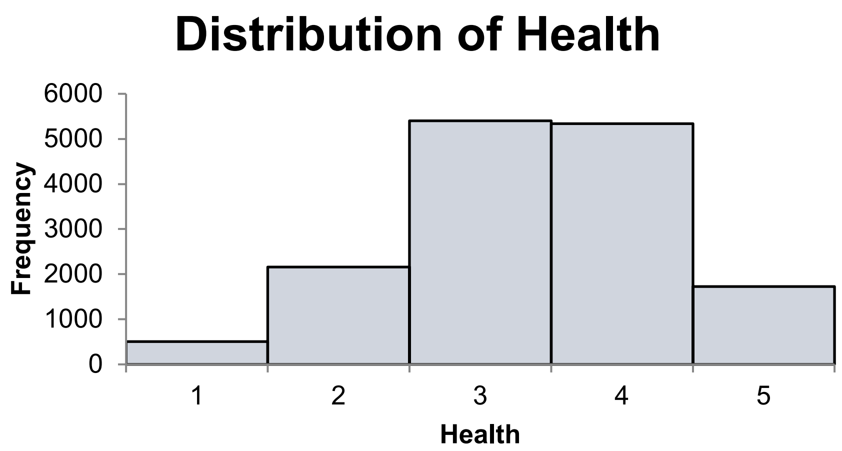


Task: Illustrate the distribution of: BMI; Health

Excel

Pivot Table

Excel Data Analysis Package Pivot Table

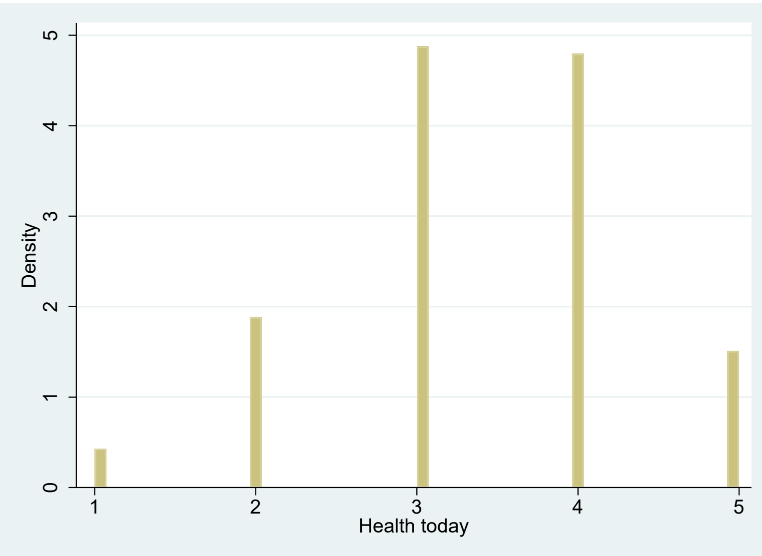


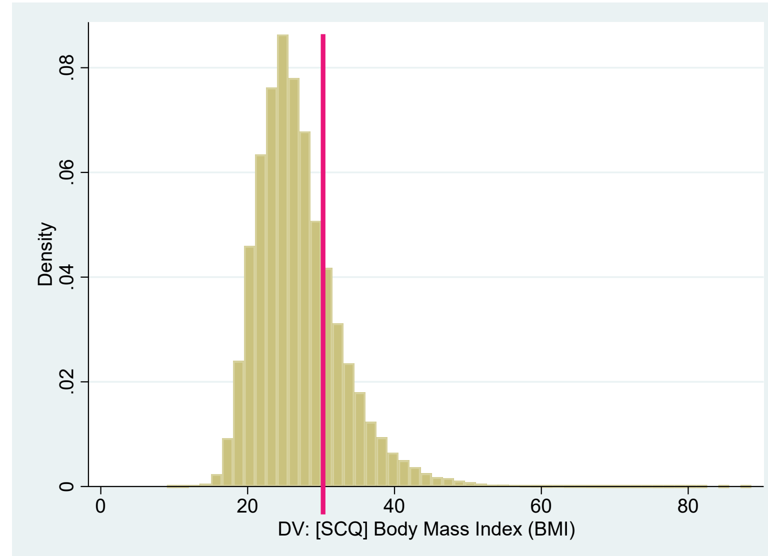
Task: Illustrate the distribution of: a. BMI; b. Health

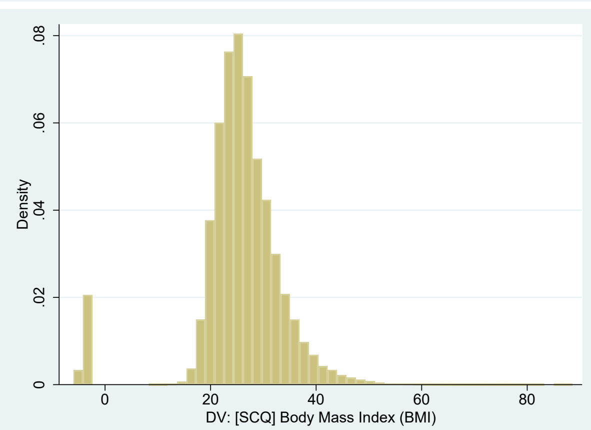
Excel

Distribution of Heath & Obesity

obese defined as a BMI 30 or greater (Blyth, Frijters & Beatton, 2023)





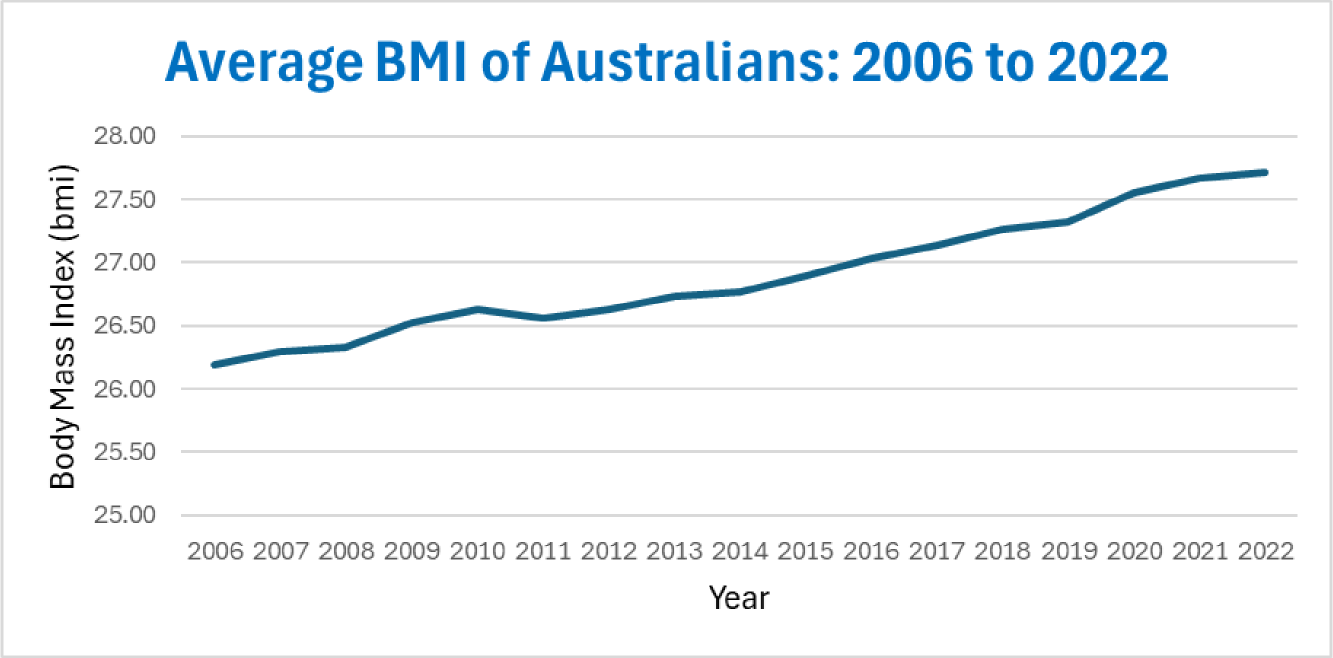


DV: [SCQ] Body Mass Index (BMI)	Freq.	Percent	Cum.
[-6.0] Implausible value	1,427	0.59	0.59
[-4.0] Refused/Not stated	8,863	3.68	4.27
9	1	0.00	4.27
9.6	1	0.00	4.27

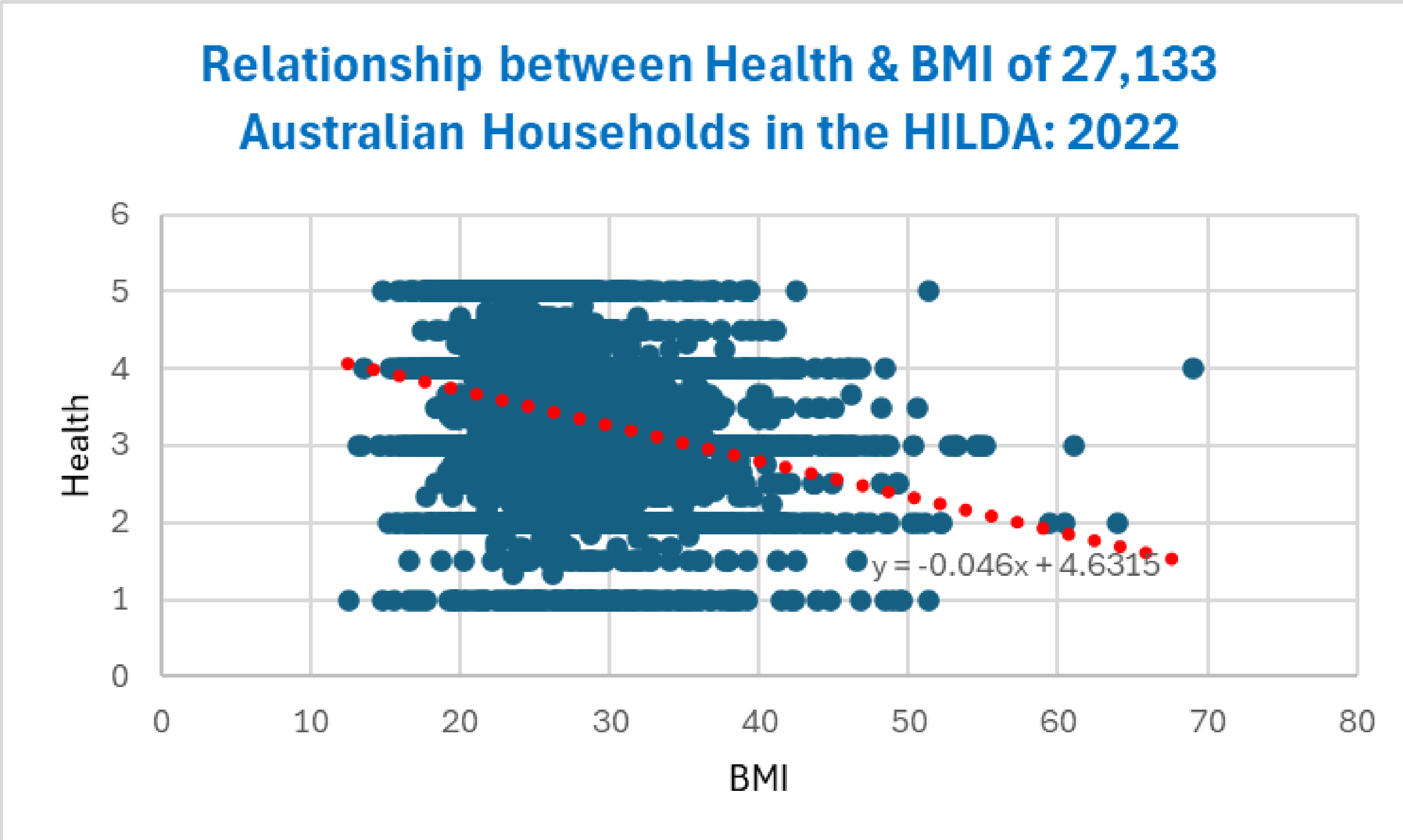
Distribution of a Variable is Determined by its Values.

6. Graphically represent the Health – BMI relationship and interpret: Time-series

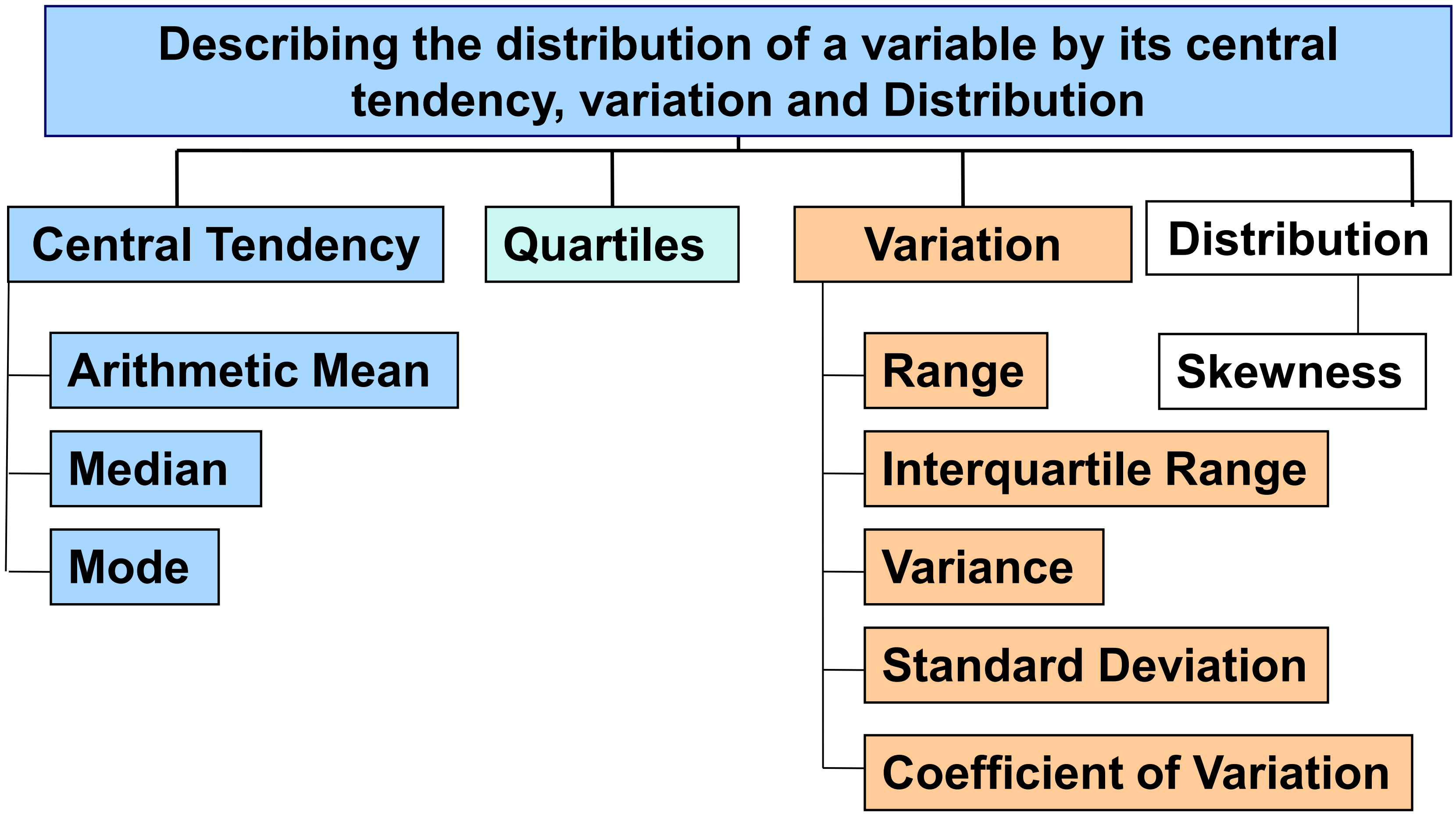
Year	mean(bmi)
2006	26.19
2007	26.30
2008	26.33
2009	26.52
2010	26.63
2011	26.56
2012	26.63
2013	26.73
2014	26.77
2015	26.89
2016	27.03
2017	27.14
2018	27.26
2019	27.33
2020	27.55
2021	27.67
2022	27.72



6. Graphically represent the Health – BMI relationship and interpret: Cross-section.



Statistics that describe the Distribution of a Variable



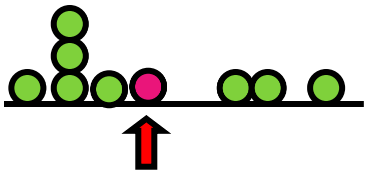
Measures of Central Tendency

Central Tendency

Arithmetic Mean

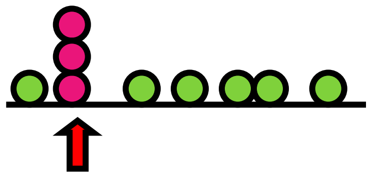
$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

Median



Midpoint of ranked values

Mode



Most frequently observed value

[Excel]

Arithmetic Mean

For sample of size n the sample mean is denoted $\overline{X}$ is calculated:

$$\overline{X} = \frac{\sum_{i=1}^{n} X_i}{n} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

where Σ means "sum" or "add up"

X_is are all observed values

Task: Some are scared by mathematical formulas;
How can we make them easy to understand?

Excel

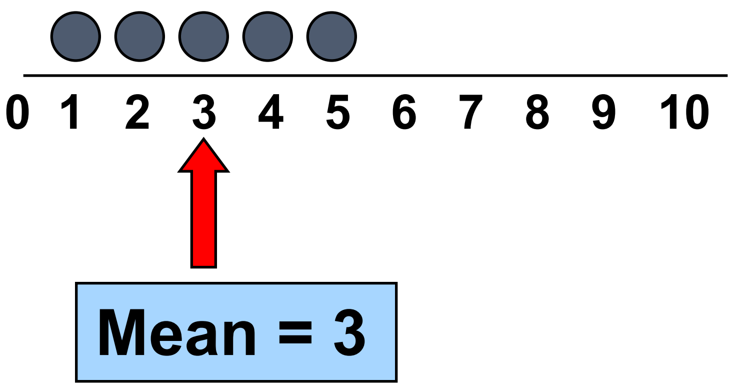
Temp (x)	Soft Drinks (y)
14	60
22	40
28	100
30	100
36	75

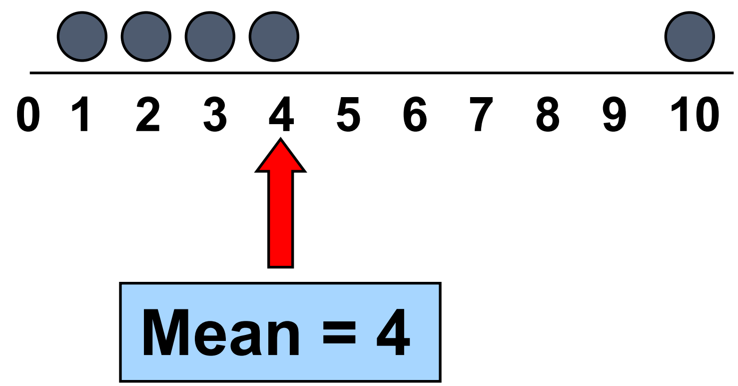
Average = $S_x = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$ = **26.00** **=AVG(A3:A7)**

Excel

Arithmetic Mean & Outliers

- The most common measure of central tendency (typically referred to as the mean)
- Commonly called the average
- Calculated as the sum of values divided by the number of values
- Affected by extreme values (outliers)



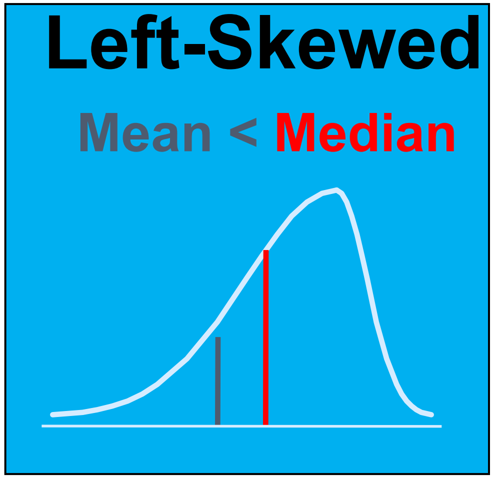


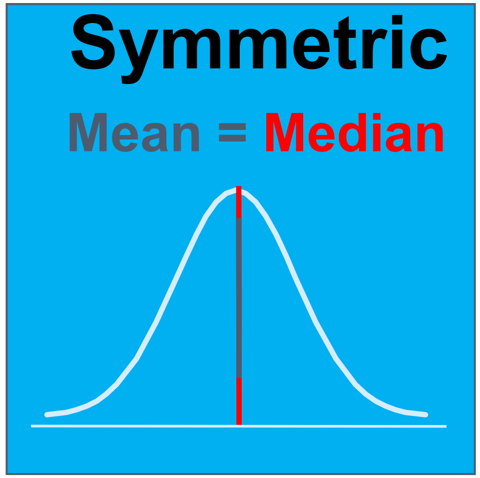
$$\frac{1+2+3+4+5}{5} = \frac{15}{5} = 3$$

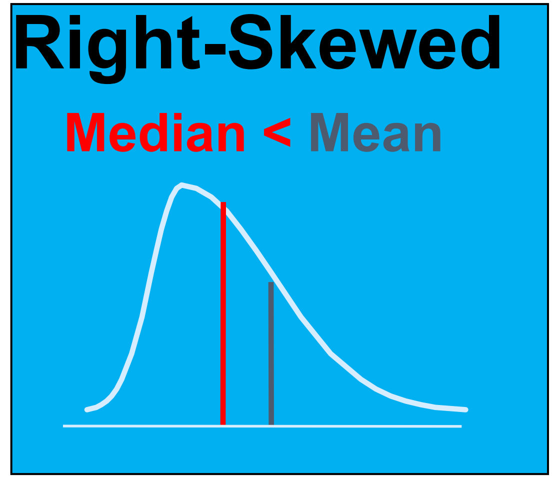
$$\frac{1+2+3+4+10}{5} = \frac{20}{5} = 4$$

Outliers

Recall: Shape(s) of a Distribution







Outliers to the left?

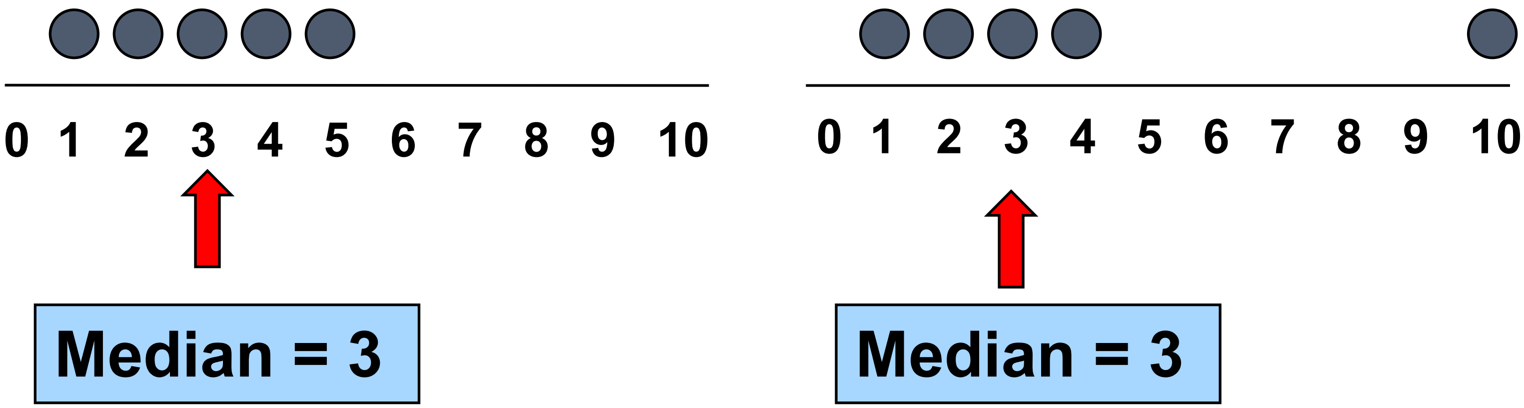
No Outliers

Outliers to the right?

Task: Why would a distribution be shaped like this?

Median

- In an ordered array, the median is the 'middle' number
  - 50% above, 50% below



- Not affected by extreme values

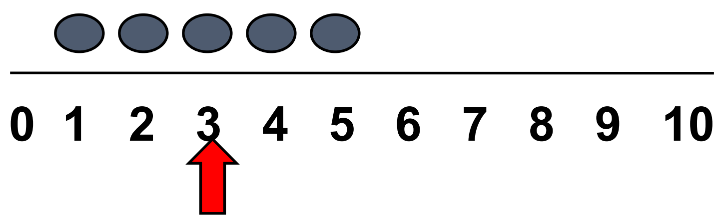
Finding the Median

- The location of the median:

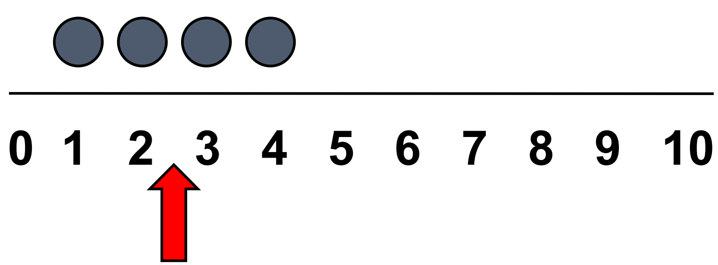
$$Median = \frac{n+1}{2} \ ranked\ value$$

- Note that $\frac{n+1}{2}$ is not the *value* of the median, only the *position* of the median in the ranked data

- Rule 1: If the number of values in the data set is odd, the median is the middle ranked value



- Rule 2: If the number of values in the data set is even, the median is the mean (average) of the two middle ranked values



$\frac{2+3}{2} = 2.5$

Excel

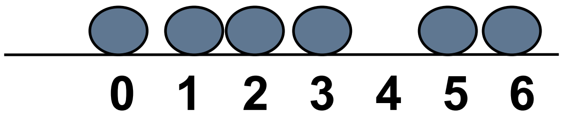
Temp (x)	Soft Drinks (y)
14	60
22	40
28	100
30	100
36	75

$$Median = \frac{n+1}{2}\ ranked\ value$$ = **28.00** **=MEDIAN(A3:A7)**

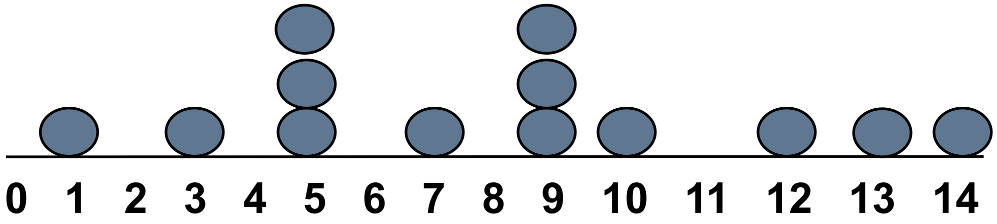
Excel

Mode

- A measure of central tendency
- Value that occurs most often (the most frequent)
- Not affected by extreme values
- Unlike mean and median, there may be no unique (single) mode for a given data set
- Used for either numerical or categorical (nominal) data
- There may may be no mode



- There may be several modes



Modes = 5 and 9

Excel

Temp (x)	Soft Drinks (y)
14	60
22	40
28	100
30	100
36	75

Mode (Soft Drinks)	=	**100.00**	**=MODE(B3:B7)**

[Excel]

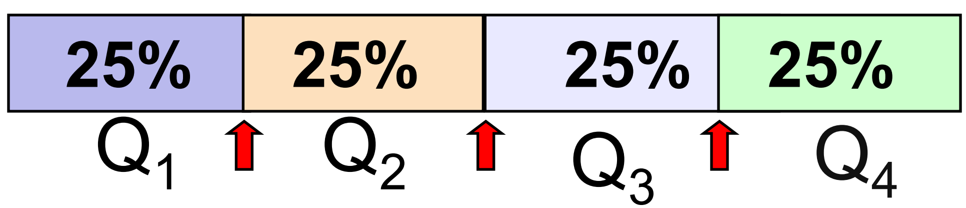
Quartiles

Quartiles split the ranked data into 4 segments with an equal number of values per segment

The first quartile, Q_1, is the value for which 25% of the observations are smaller and 75% are larger

The second quartile, Q_2, is the same as the median (50% are smaller, 50% are larger)

Only 25% of the observations are greater than the third quartile Q_3



Q1 and Q3 are measures of non-central location

Q2 = median, a measure of central tendency

Quartile Formulas

Find a quartile by determining the value in the appropriate position in the *ranked* data, where

First quartile position: **$Q_1 = (n+1)/4$**

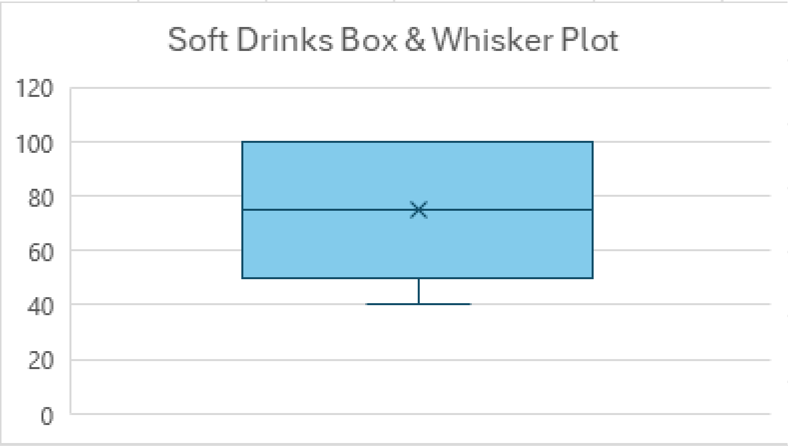
Second quartile position: **$Q_2 = (n+1)/2$** (the median)

Third quartile position: **$Q_3 = 3(n+1)/4$**

where **n** is the number of observed values

Excel

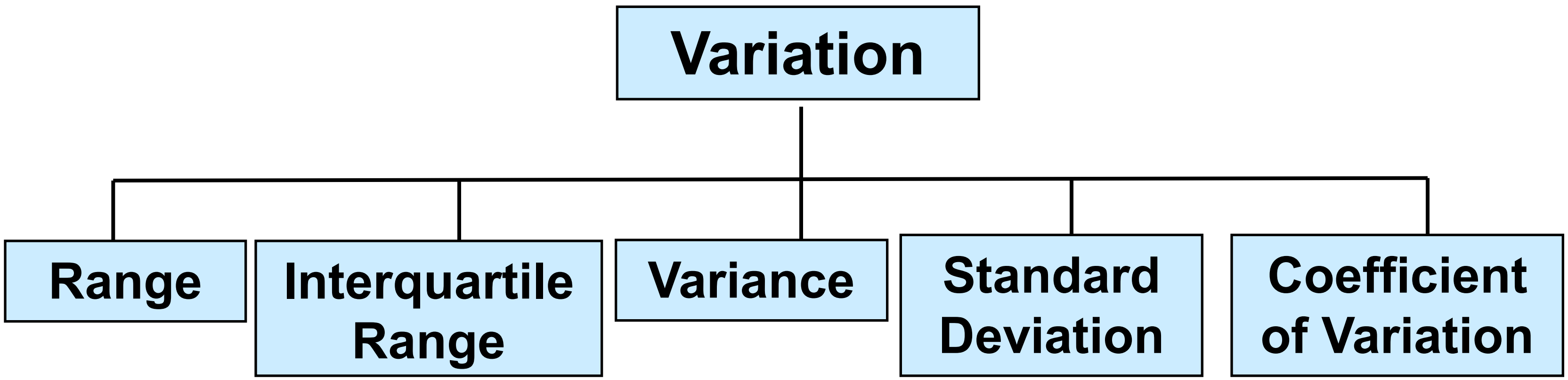
Temp (x)	Soft Drinks (y)
14	60
22	40
28	100
30	100
36	75

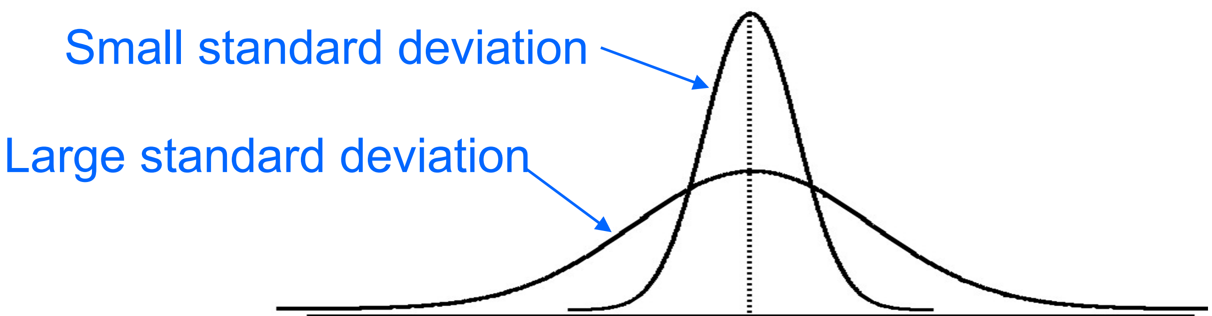


Quartile (Soft Drinks), 4	=	**100**	**=QUARTILE(B3:B7,4)**

Excel

Statistics that describe Variation in the Distribution of a Variable





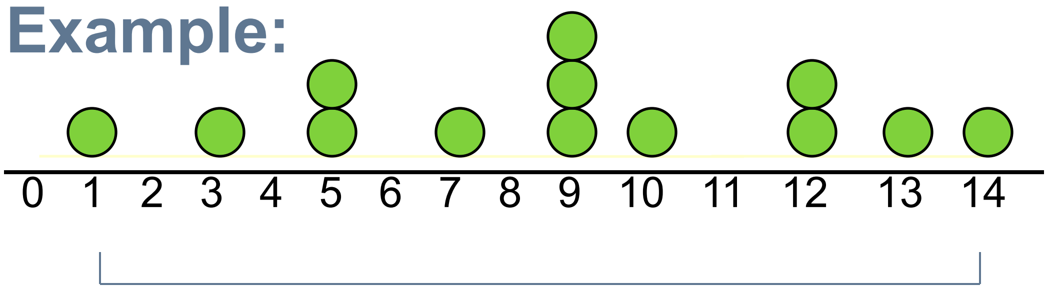
Click here to view Variation in a Distribution

Range

- Simplest measure of variation
- Difference between the largest and the smallest values in a set of data

$$\text{Range} = X_{largest} - X_{smallest}$$

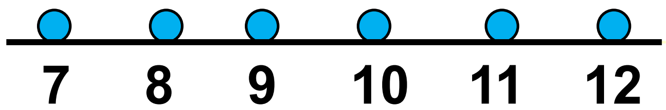
Example:



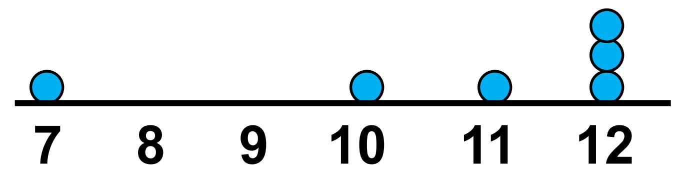
Range = 14 - 1 = 13

Disadvantages of the Range

- Ignores the distribution of the data



Range = 12 - 7 = 5



Range = 12 - 7 = 5

- Like the mean, range is sensitive to outliers

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,3,3,3,3,4,5

Range = 5 - 1 = 4

1,1,1,1,1,1,1,1,1,1,1,2,2,2,2,2,2,2,3,3,3,3,4,120

Range = 120 - 1 = 119

Interquartile Range (IQR)

- Like the median and Q_1 and Q_2, the IQR is a resistant summary measure

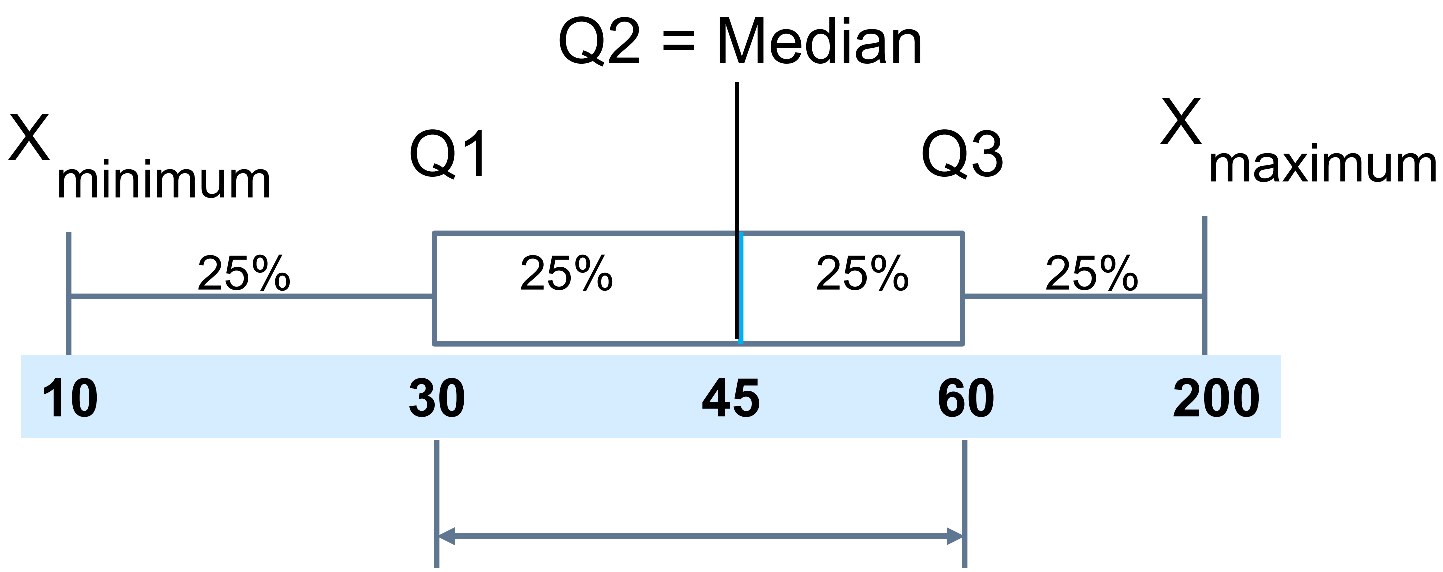
- Eliminates outlier problems by using the **interquartile range** as high- and low-valued observations are removed from calculations

- IQR = 3rd quartile – 1st quartile

$$\boxed{Q_3 - Q_1}$$

Interquartile Range

Example: Range = 200 – 10 = 190 (Misleading)



IQR = 60 – 30 = 30

Even if the value of 200 changes to 300, IQR remains the same, hence resistant to changes in extreme values

Coefficient of Variation

- Measures relative variation i.e. shows variation relative to mean

- Can be used to compare two or more sets of data measured in different units

- Always expressed as percentage (%)

$$\mathrm{CV} = \left(\frac{\mathrm{S}}{\overline{\overline{\mathrm{X}}}}\right) * 100\%$$

Excel

Temp (x)	Soft Drinks (y)
14	$ 60.00
22	$ 40.00
28	$ 100.00
30	$ 100.00
36	$ 75.00

$$CV = \left(\frac{S}{\overline{X}}\right) * 100\%$$

Coefficient of Variation (Temp) =	=	**32.18**	**%**	**=STDEV(A3:A7)/=AVG(A3:A7)**
Coefficient of Variation (Soft Drinks) =	=	**34.64**	**%**	**=STDEV(B3BA7)/=AVG(B3:B7)**

Dispersion of Variables about their Averages

- Population
- Sample

Dispersion of the variables about their averages: Variance

- Mean squared deviation

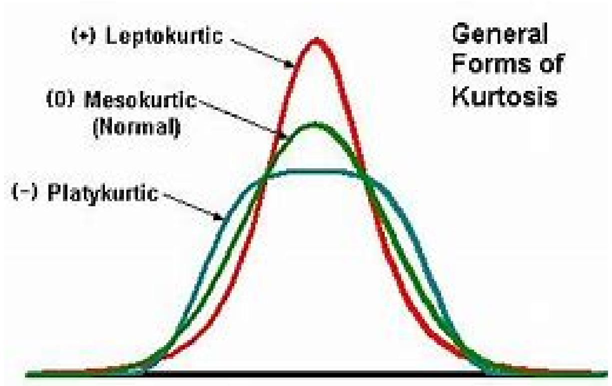
Sample Variance

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

The denominator (*n-1*) is to adjust for the biasness of the sample statistics.



Task: Why are the variance formulas different?

Variance Example

Stock A	0	2	5	5	5	6	6	7	8	9
Stock B	4	4	5	5	5	5	6	6	6	7

Calculate the sample standard deviation for stocks A and B

$$s_A{}^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

$$= \frac{(0-5.3)^2 + (2-5.3)^2 + \cdots + (9-5.3)^2}{10-1}$$

$$= \frac{64.1}{9}$$

$$= 7.12$$

$$s_A = \sqrt{7.12}$$

$$= 2.67$$

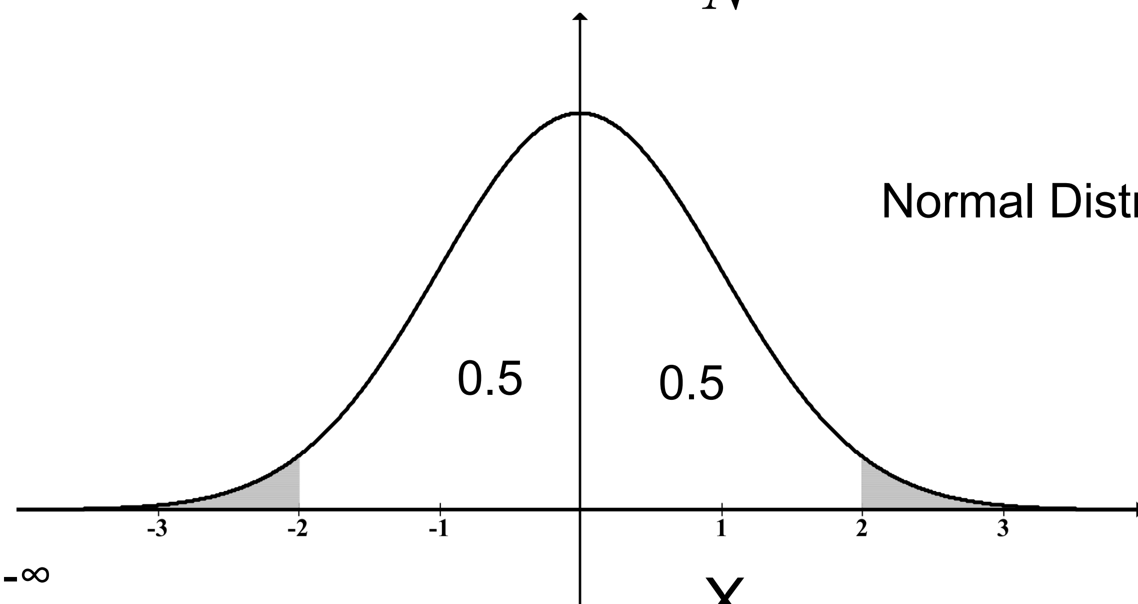
Excel Data Analysis Toolpack

Standard Deviation - Population

Most commonly used measure of variation; Shows variation about the mean; has the same units as the original data

$$6 = \sqrt{6^2}; \quad \sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N} \quad ; (+0 \ldots .+\infty)$$

Normal Distribution ~N (0,1)



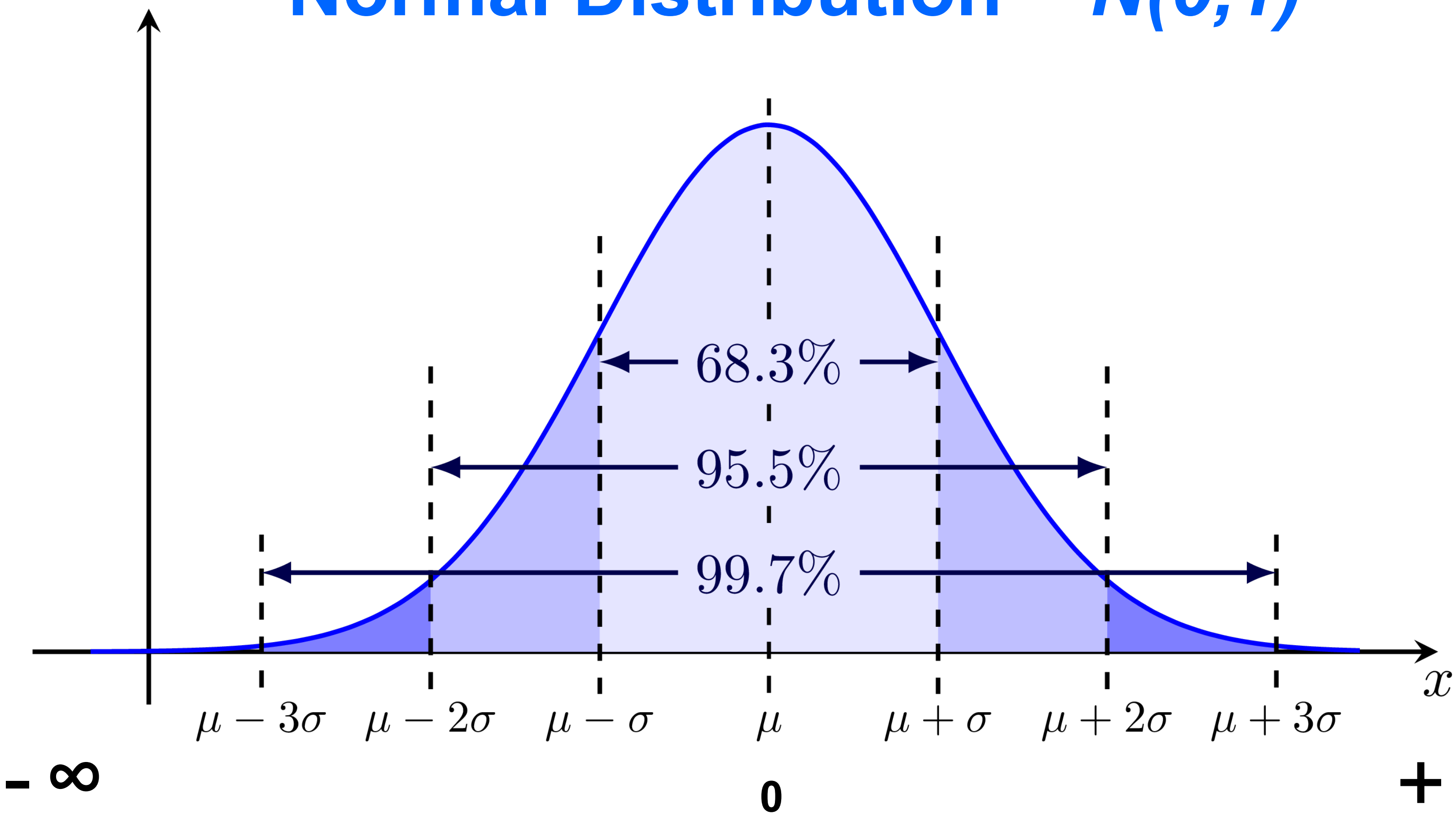
Max: -∞

Max: +∞

X_1

$\mu = 0$

Dispersion of X values in a Normal Distribution ~ *N(0,1)*



e.g. 68.3% of observations lie within +/- 1 Standard Deviation, either side, of the mean

Standard Deviation - Sample

Most commonly used measure of variation; Shows variation about the mean; has the same units as the original data

$$S = \sqrt{S^2}; \quad S_x = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$$

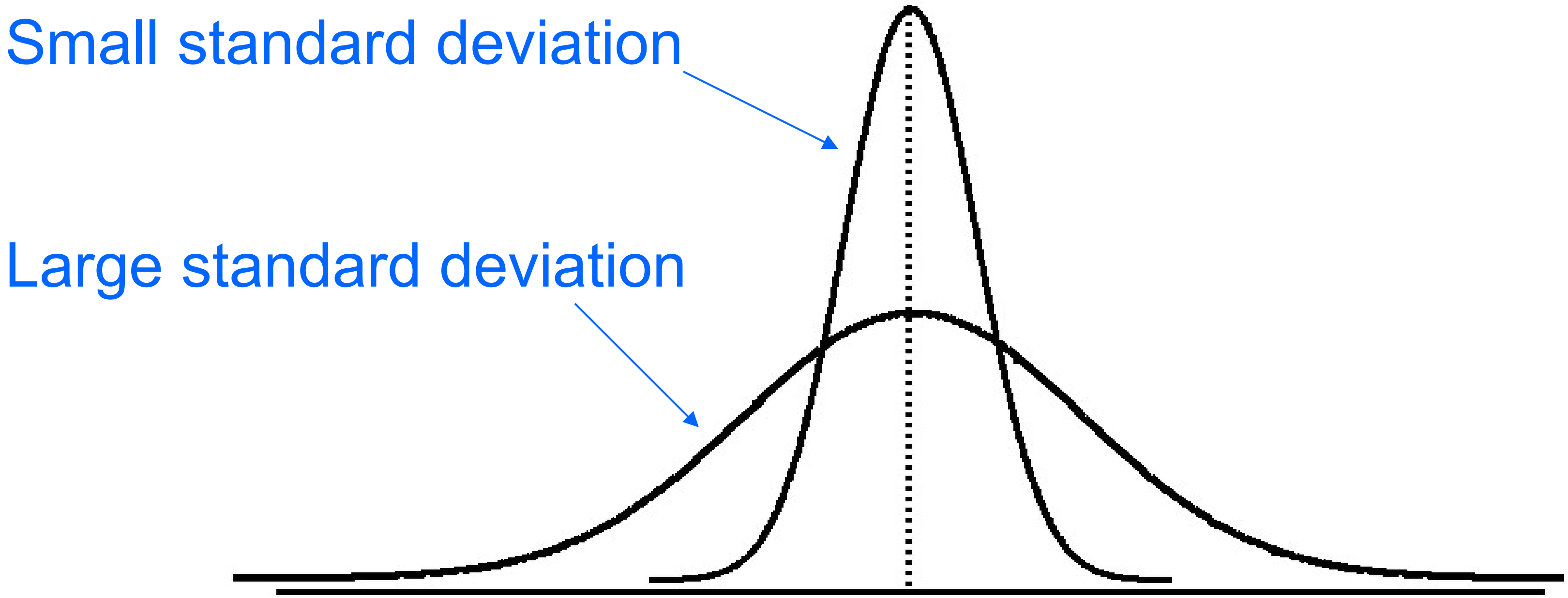
Excel

Temp (x)	Soft Drinks (y)
14	60
22	40
28	100
30	100
36	75

Standard Deviation = $S_x = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \overline{X})^2}{n-1}}$ = **8.37** **=STDEV.S(A3:A7)**

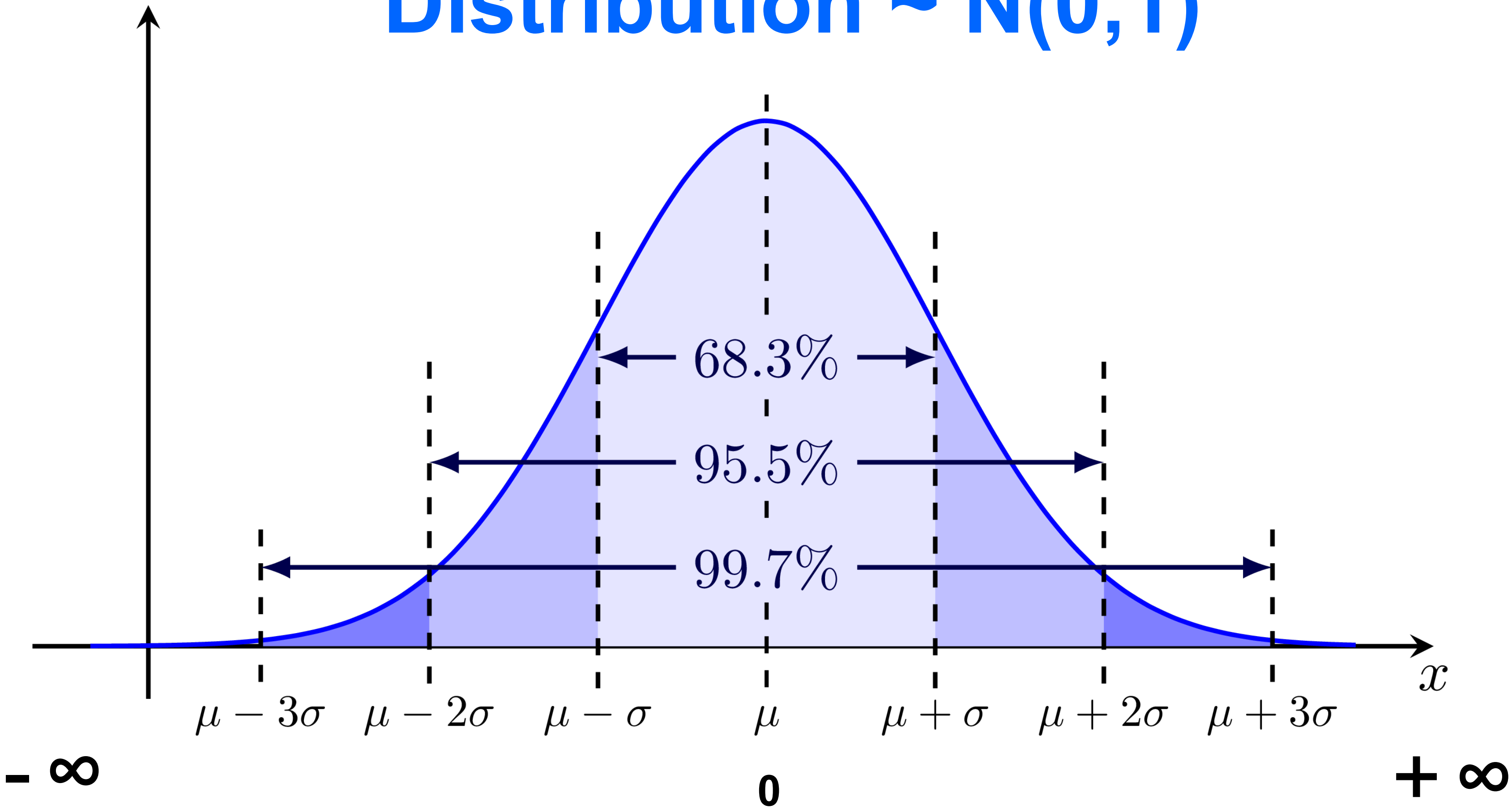
Excel

Visualize a distribution



Continuous Probability Distributions

Dispersion of X values in a Normal Distribution ~ N(0,1)



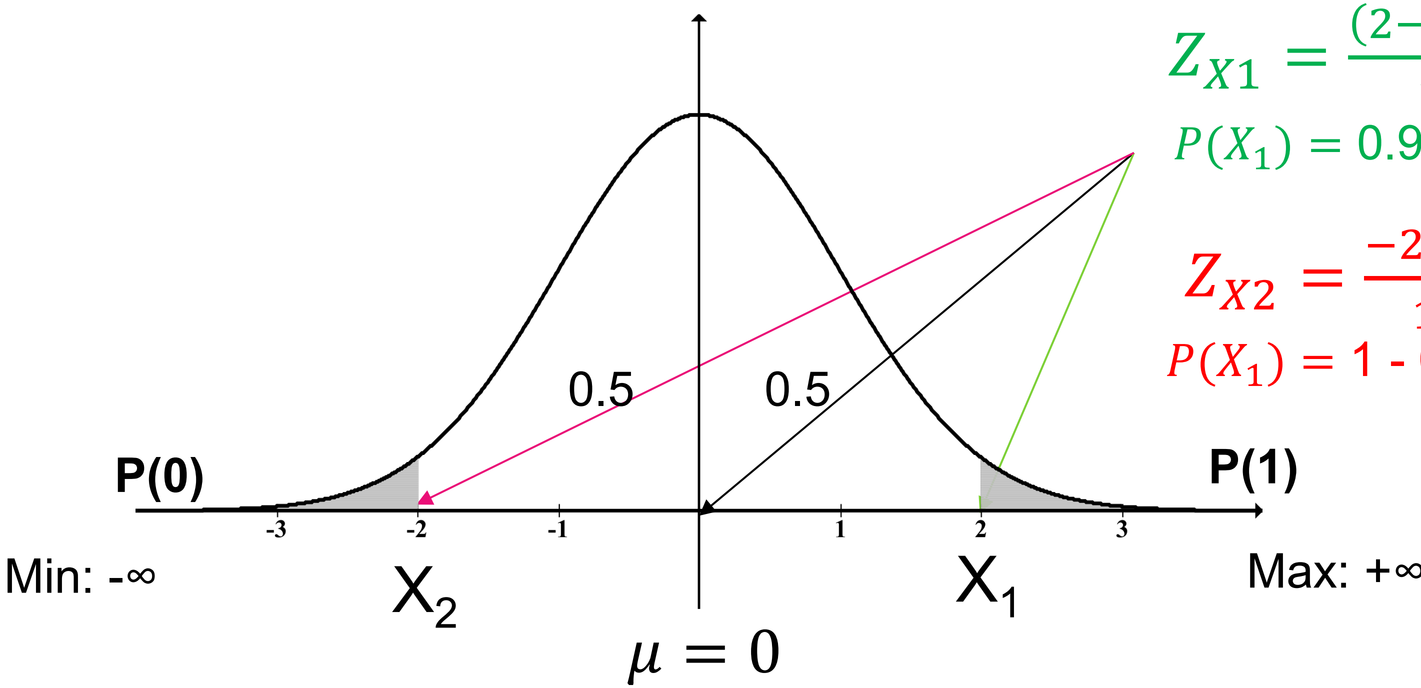
e.g. 68.3% of observations lie within +/- 1 Standard Deviation, either side, of the mean

Distance from the Mean - Population

We can also express the position of a value of x relative to its mean in terms of a **continuous probability**

Normal Distribution: ~N(0, 1)

$$Z = \frac{(x_i - \mu)}{\sigma}$$



$$Z_{X1} = \frac{(2-0)}{1} = 2$$

$P(X_1) = 0.9772$

$$Z_{X2} = \frac{-2-0}{1} = -2$$

$P(X_1) = 1 - 0.9772 = 0.0228$

[Z Tables]

Next Geek 2: Relationship Statistics

End of Session